Ohm's law states that, *"At a constant* temperature *in any closed circuit, the current is directly proportional to the voltage applied and inversely proportional to the resistance of the circuit."*

i.e $I \alpha V$ and $I \alpha 1/R$ from this

$V = I R$ and $R = \frac{V}{I}$

Where, V = Voltage in volts. I = current in ampere and R = Resistance in ohm



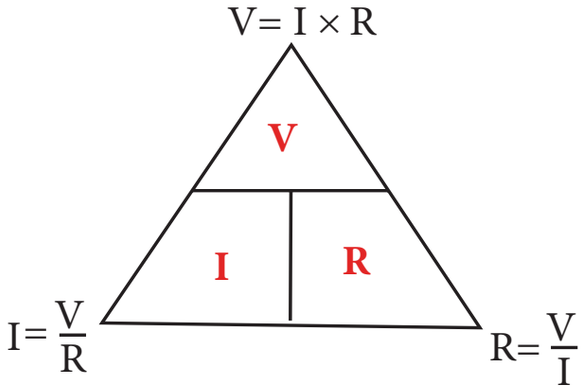
Georg Simon Ohm

Born: 1789

Died: 1854

Georg Simon Ohm (1789 – 1854) was a German physicist and mathematician. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist Alessandro Volta.

Ohm found that there is a direct proportionality between the potential difference applied across a conductor and the resultant electric current. This relationship is known as **Ohm's law**.



Problems

1) *Supply voltage of the circuit is 240V. The resistance value is 60Ω. Calculate the current flowing through this circuit.*

Voltage (V) = 240V
Resistance (R) = 60Ω
Current (I) = ?
According to ohm's law, $I = \frac{V}{R}$

Current $I = \frac{240}{60} = 4A$

2) *An electrical circuit has 1000W power and the value of voltage is 240V. Find the value of current flowing through it.*

Power (P) = 1000W
Voltage(V) = 240V
Current(I) = ?
Power, $P = V \times I$
$1000 = 240 \times I$
$I = \frac{1000}{240} = 4.16$ A

3) *Voltage of the circuit is 230V and current 10A is flowing through it. Find the value of Resistance.*

Voltage (V) = 230V
Current (I) = 10 A
Resistance (R) =?
According to ohm's law, $I = \frac{V}{R}$

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